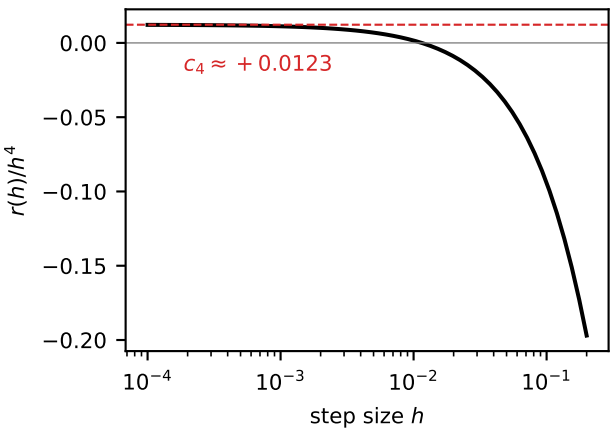
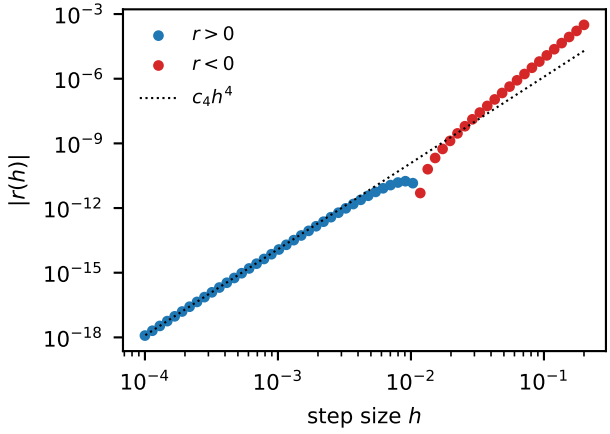

(a) signed RK4 energy defect
$c_4 \approx +0.0123$
$r(h)/h^4$
0.00
−0.05
−0.10
−0.15
−0.20
10^{-4}
10^{-3}
10^{-2}
10^{-1}
step size h


(b) sign change near $h \approx 0.011$
$r > 0$
$r < 0$
$c_4 h^4$
$|r(h)|$
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
step size h